PCCST501 - COMPUTER NETWORKS

Assignment 3 | Module 3

LAN, Ethernet & Wireless Network Investigation

Data-Link Layer, Wireless LANs & Mobile IP

Course Outcome	CO3 - Analyze how data-link protocols provide node-to-node delivery, framing, addressing and access to shared media, with emphasis on Ethernet and wireless networking.
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Deliverables	PDF report, Ethernet frame diagram, Wireshark captures, wireless LAN design, comparison table, README.md

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Q1. Responsibilities of the Data-Link Layer; LLC vs MAC

The data-link layer (Layer 2 of the OSI model) moves a datagram from one node to the **adjacent** node over a single link (hop-by-hop, node-to-node delivery). A route across many networks is the job of the network layer; the data-link layer only guarantees correct transfer over one link, which may be wired (Ethernet) or wireless (Wi-Fi). It uses the physical layer's raw bit stream and offers a frame-based service to the network layer.

Main responsibilities

- **Framing** - encapsulates the network-layer packet with a header and trailer so the receiver can find where a frame starts and ends (preamble/SFD in Ethernet, PLCP preamble in Wi-Fi).
- **Physical (MAC) addressing** - source and destination hardware addresses (48-bit) identify the sender and receiver on the same link; they are rewritten at every hop.
- **Node-to-node delivery** - ensures the frame reaches the next hop; the destination address lets NICs discard frames that are not meant for them.
- **Access to shared media** - when many stations share a channel (hub-based Ethernet, a Wi-Fi channel) a medium-access control protocol (CSMA/CD, CSMA/CA) decides who may transmit.
- **Error detection (and sometimes correction/retransmission)** - a CRC in the trailer (FCS) detects bit errors; corrupted frames are dropped. Wi-Fi additionally uses ACK and retransmission because wireless is lossy.
- **Flow control** - prevents a fast sender from overrunning a slow receiver (e.g., 802.3x PAUSE frames in Ethernet).
- **Link management** - association/authentication in Wi-Fi, link aggregation and VLAN tagging in Ethernet.

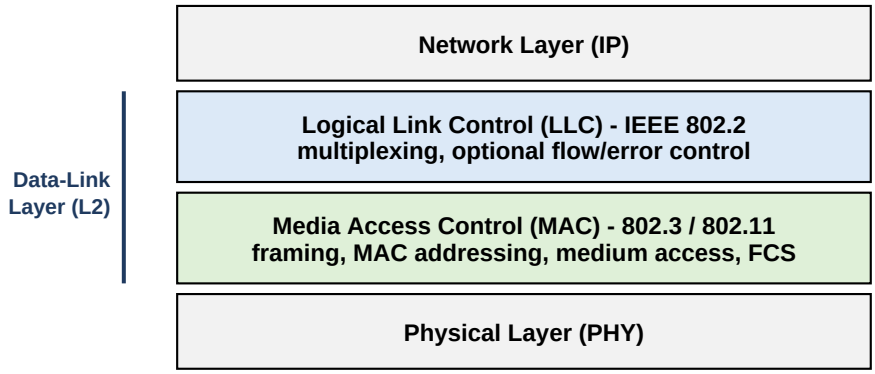


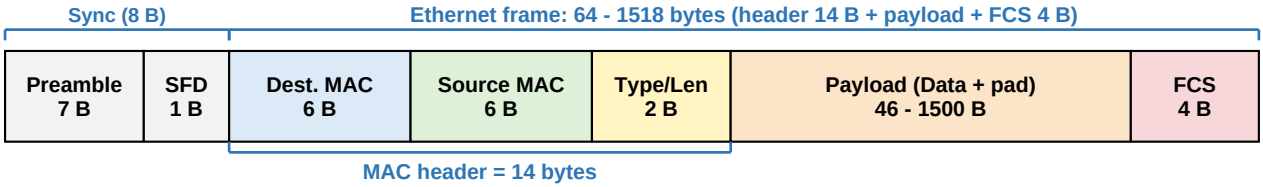
Figure 1.1 - The data-link layer split into LLC and MAC sublayers (IEEE 802 model).

LLC vs MAC

Aspect	Logical Link Control (LLC) - IEEE 802.2	Media Access Control (MAC)
Position	Upper sublayer, directly below the network layer	Lower sublayer, directly above the physical layer
Main job	Interface to the network layer; identifies which upper-layer protocol owns the data; optional flow and error control	Framing, MAC addressing, controlling access to the shared medium, error detection (FCS)
Addressing	Service Access Points (DSAP, SSAP) identify protocol entities	48-bit MAC addresses identify NICs
Medium dependence	Independent of the medium and the access method - same for all 802 LANs	Specific to technology: 802.3 (CSMA/CD), 802.11 (CSMA/CA), 802.5, etc.
Services	Type 1 connectionless unacknowledged (most used), Type 2 connection-oriented, Type 3 acknowledged connectionless	Best-effort frame delivery; ACK only in 802.11
Header	3-4 bytes: DSAP, SSAP, Control (SNAP header adds OUI + EtherType)	Ethernet 14 B header + 4 B FCS; 802.11 24-30 B header + 4 B FCS
In practice	Ethernet II (DIX) frames use the Type field instead of LLC, so LLC is often invisible; 802.11 still carries LLC/SNAP	Implemented in NIC hardware / firmware

Q2. Ethernet Frame Format

Ethernet (IEEE 802.3 / Ethernet II) transmits data as frames. The format below is Ethernet II, the one used for IP traffic. The first 8 bytes (preamble + SFD) are consumed by the physical layer for clock synchronisation and are not counted in the frame size.



Inter-Frame Gap (IFG): 96 bit times (12 B) of idle before the next preamble.
Type/Len >= 0x0600 (1536) = EtherType (Ethernet II); <= 1500 = payload length (IEEE 802.3 + LLC).

Figure 2.1 - Ethernet II frame diagram.

Field	Size	Purpose
Preamble	7 B	Alternating 10101010... pattern that lets the receiver synchronise its clock with the sender.
SFD (Start Frame Delimiter)	1 B	10101011 - marks the end of the preamble; the next byte is the first byte of the frame.
Destination MAC	6 B	Hardware address of the intended receiver (unicast, multicast or broadcast FF:FF:FF:FF:FF:FF). Each NIC compares it with its own address and drops non-matching frames. Bit 0 of the first byte is the I/G bit (0 = individual, 1 = group); bit 1 is the U/L bit (0 = globally unique, vendor-assigned).
Source MAC	6 B	Hardware address of the sending NIC. Always unicast. Switches read it to learn which port a host sits behind. First 3 bytes = OUI (vendor), last 3 = NIC-specific.
Type / Length	2 B	If value >= 0x0600 it is an EtherType naming the upper-layer protocol (0x0800 IPv4, 0x0806 ARP, 0x86DD IPv6, 0x8100 802.1Q VLAN tag). If <= 1500 it is the payload length (IEEE 802.3 with LLC).
Payload / Data	46-1500 B	The network-layer packet. Maximum Transmission Unit (MTU) = 1500 B. If data is less than 46 B, padding is added so the frame is at least 64 B (needed for collision detection).
FCS (Frame Check Sequence)	4 B	CRC-32 computed over destination, source, type and payload. The receiver recomputes it; a mismatch means corruption and the frame is silently discarded (no retransmission at Layer 2).

Frame size limits: minimum 64 B and maximum 1518 B (1522 B with a 4-byte 802.1Q VLAN tag) measured from destination MAC to FCS. Including preamble/SFD and the 12-byte inter-frame gap, the smallest frame occupies 84 B on the wire.

Example (frame header in hex):

```
00 1a 2b 3c 4d 5e | 08 00 27 aa bb cc | 08 00 | 45 00 00 54 ..... | (4-byte FCS)
Dest MAC          Source MAC          Type    Payload (IPv4 header + data)
Type 0x0800 -> IPv4;  Dest I/G bit = 0 (unicast);  Source OUI 08:00:27 (VirtualBox virtual NIC)
```

Q3. Wireshark Capture - Source and Destination MAC Addresses

Procedure

- Connect the host to a wired Ethernet port (or a virtual NIC bridged to it) and open Wireshark. Select the Ethernet interface (e.g., `eth0` / `Ethernet`) and press Start.
- Generate traffic: `ping 192.168.1.1`, `nslookup example.com` and open a web page.
- Use the display filter `eth.addr == 08:00:27:aa:bb:cc` or `arp || icmp || dns`.
- Add columns for `eth.src` and `eth.dst` (right-click the field in the details pane, *Apply as Column*).
- Stop capture, save as `captures/ethernet_capture.pcapng` and take a screenshot of the packet list and the expanded *Ethernet II* tree.
- Cross-check with *Statistics* > *Endpoints* > *Ethernet*.

Captured frames (packet list, with Ethernet columns)

No.	Time (s)	Source MAC (eth.src)	Destination MAC (eth.dst)	Protocol	Length	Info
1	0.000000	08:00:27:aa:bb:cc	ff:ff:ff:ff:ff:ff	ARP	42	Who has 192.168.1.1? Tell 192.168.1.10
2	0.000412	00:1a:2b:3c:4d:5e	08:00:27:aa:bb:cc	ARP	60	192.168.1.1 is at 00:1a:2b:3c:4d:5e
3	0.001020	08:00:27:aa:bb:cc	00:1a:2b:3c:4d:5e	ICMP	98	Echo (ping) request id=0x0001 seq=1
4	0.003880	00:1a:2b:3c:4d:5e	08:00:27:aa:bb:cc	ICMP	98	Echo (ping) reply id=0x0001 seq=1
5	0.512300	08:00:27:aa:bb:cc	00:1a:2b:3c:4d:5e	DNS	74	Standard query A example.com (to 192.168.1.1)
6	0.830100	08:00:27:aa:bb:cc	01:00:5e:00:00:fb	MDNS	82	Standard query PTR _services._dns-sd._udp.local

Figure 3.1 - Representative Wireshark packet list (six frames; the five required plus one multicast). MAC addresses are examples.

Expanded Ethernet II header of frame 3

```
Frame 3: 98 bytes on wire (784 bits), 98 bytes captured
Ethernet II, Src: 08:00:27:aa:bb:cc, Dst: 00:1a:2b:3c:4d:5e
    Destination: 00:1a:2b:3c:4d:5e
        .... ..0. .... .... .... .... = LG bit: Globally unique address
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source:      08:00:27:aa:bb:cc   (OUI 08:00:27 = VirtualBox virtual NIC)
    Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.1
Internet Control Message Protocol  Type: 8 (Echo (ping) request)

0000  00 1a 2b 3c 4d 5e 08 00  27 aa bb cc 08 00 45 00   ..+<M^.. '.....E.
0010  00 54 3a 1c 40 00 40 01  ..                         .T:.@.@. ..
```

Analysis

- **Frame 1** is an ARP request: the destination MAC is the broadcast address `ff:ff:ff:ff:ff:ff` because the sender does not yet know the gateway's MAC. Every host on the segment receives it.
- **Frame 2** is the ARP reply, sent as **unicast** directly back to the requester; it carries the gateway MAC 00:1a:2b:3c:4d:5e.
- **Frames 3-4** show a ping to the gateway: the source and destination MACs simply swap between request and reply.
- **Frame 5** is a DNS query to an external name. Even though the final server is on the Internet, the Ethernet destination is the **default gateway's MAC** - MAC addresses only identify the next hop, while the IP address identifies the end host.
- **Frame 6** uses a multicast destination: 01:00:5e is the IPv4-multicast MAC prefix (224.0.0.251 maps to 01:00:5e:00:00:fb).

Q4. Switch Operation and MAC Address Learning

A switch is a Layer-2 device that forwards frames based on the destination MAC address. It maintains a **MAC address table** (also called CAM table or forwarding table) mapping *MAC address -> port (+ age)*. The table is built automatically by **backward learning**; no configuration is needed.

The learning / forwarding algorithm

- **Learn:** for every frame that arrives, read the *source* MAC and record (or refresh) it against the ingress port with an ageing timer (default 300 s on many switches).
- **Lookup:** search the table for the *destination* MAC.
- **Forward:** if found on a different port - forward only to that port (**filter/forward**); if found on the same port as the ingress - discard (**filter**); if not found (unknown unicast), broadcast or multicast - **flood** to all ports except the ingress port.
- **Age out:** entries not refreshed within the ageing time are deleted, which also handles hosts that move to another port.

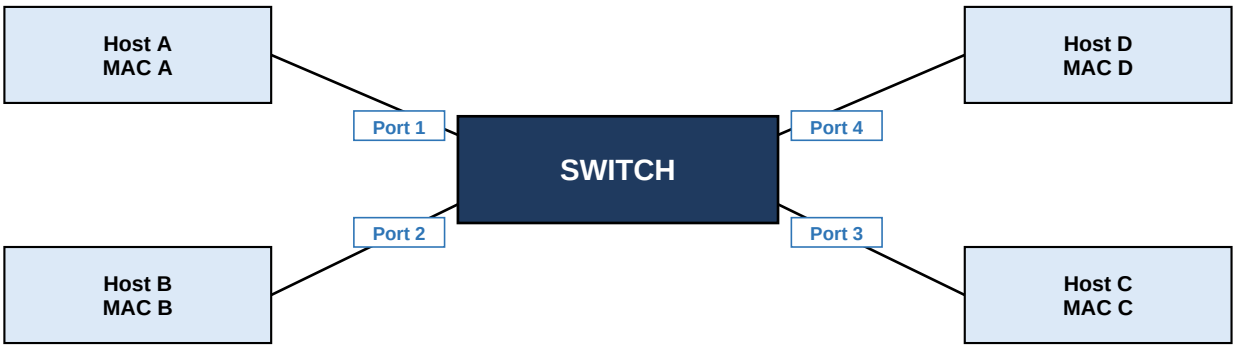


Figure 4.1 - Demonstration topology: four hosts (A-D) on ports 1-4 of a switch. The table starts empty.

Building the forwarding table, step by step

Step	Frame (Src -> Dst)	Learning (Src)	Lookup (Dst)	Action	Table afterwards
0	-	-	-	Switch powered on	(empty)
1	A -> B	A is on port 1: add	B not in table	FLOOD to ports 2, 3, 4	A:1
2	B -> A	B is on port 2: add	A found: port 1	Forward to port 1 only	A:1, B:2
3	C -> A	C is on port 3: add	A found: port 1	Forward to port 1 only (B and D do not see it)	A:1, B:2, C:3
4	A -> D	A already known: refresh timer	D not in table	FLOOD to ports 2, 3, 4	A:1, B:2, C:3
5	D -> A	D is on port 4: add	A found: port 1	Forward to port 1 only	A:1, B:2, C:3, D:4
6	A -> FF:FF:FF:FF:FF:FF (ARP)	A refreshed	Broadcast	FLOOD to ports 2, 3, 4	unchanged
7	B -> C	B refreshed	C found: port 3	Forward to port 3 only	unchanged

Final forwarding table (steps 0-7)

MAC address	Port	Type	Ageing timer
00:00:00:00:00:0A (Host A)	1	Dynamic	300 s (reset on each frame from A)
00:00:00:00:00:0B (Host B)	2	Dynamic	300 s
00:00:00:00:00:0C (Host C)	3	Dynamic	300 s
00:00:00:00:00:0D (Host D)	4	Dynamic	300 s

On a real Cisco switch the table can be observed (and cleared to repeat the experiment) with:

```
Switch# clear mac address-table dynamic
Switch# show mac address-table dynamic
          Mac Address Table
-------------------------------------------
Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
   1    0000.0000.000a    DYNAMIC     Fa0/1
   1    0000.0000.000b    DYNAMIC     Fa0/2
   1    0000.0000.000c    DYNAMIC     Fa0/3
   1    0000.0000.000d    DYNAMIC     Fa0/4
```

Benefit: after learning, unicast traffic is confined to the two ports involved, so bandwidth is not wasted and other hosts cannot passively see it (they could on a hub). Note that flooding of unknown unicast and a MAC-flooding attack (CAM overflow) are the reasons port security is configured on access ports.

Q5. Hub vs Switch vs Bridge

Feature	Hub	Bridge	Switch
OSI layer	Layer 1 (physical)	Layer 2 (data link)	Layer 2 (some models Layer 3)
Intelligence	None - repeats electrical signal / bits	Reads MAC addresses, learns, filters	Same as bridge, in hardware (ASIC)
Forwarding	Broadcasts every incoming bit to all other ports	Forwards frames only to the segment holding the destination (store-and-forward)	Forwards to the destination port only; store-and-forward, cut-through or fragment-free
Collision domains	**One** collision domain for all ports	One per port/segment (typically 2-4)	**One per port** (each link is its own domain)
Broadcast domains	One	One (broadcasts are forwarded)	One per VLAN (broadcasts flood within the VLAN)
Ports	4-24	2-4 (typically)	8-48+ (also 100+ on modular chassis)
Bandwidth	Shared: total bandwidth divided among all stations	Each segment has its own bandwidth	Dedicated bandwidth per port; aggregate switching capacity (backplane)
Duplex	Half only (CSMA/CD needed)	Half or full per port	Full duplex normal; CSMA/CD not needed
Mixed speeds / media	No	Yes (e.g., Ethernet to Token Ring / Wi-Fi)	Yes - auto-negotiation 10/100/1000/10G
Extra features	None	STP, filtering	VLANs, STP/RSTP, port security, QoS, PoE, link aggregation, SPAN
Security	Poor - frames visible to all (easy sniffing)	Better	Good; supports 802.1X, port security, DAI
Practical use today	Obsolete; sometimes seen in labs or for sniffing	Mostly replaced by switches; software bridging in hypervisors, wireless-to-wired bridging, Linux bridge	Standard access/distribution device in every LAN, campus and data centre

Key points: A switch is effectively a high-speed multiport bridge implemented in hardware. Replacing a hub with a switch removes collisions and dramatically increases usable throughput without changing the NICs or cabling. Neither bridges nor switches separate broadcast domains - that requires a router (or Layer-3 switch / VLANs).

Collision domains: 8-port example

Device	Collision domains (8 hosts)	Broadcast domains
8-port hub	1	1
8-port switch	8	1
Two 4-port hubs joined by a 2-port bridge	2	1

Q6. CSMA/CD and Its Reduced Role in Modern Ethernet

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is the original MAC protocol of half-duplex Ethernet (IEEE 802.3) used on shared coax and hub-based networks. Several stations share one wire; two may transmit at nearly the same time and their signals collide.

Algorithm

- **Carrier sense:** listen to the medium. If it is busy, wait until idle (1-persistent) and then wait the inter-frame gap (96 bit times).
- **Multiple access:** if idle, start transmitting the frame.
- **Collision detection:** while transmitting, keep monitoring the line. If the received signal differs from what is being sent (or the voltage/current exceeds a threshold), a collision has occurred.
- **Jam signal:** on collision, send a 32-bit jam signal so every station notices the collision, and abort.
- **Binary exponential backoff:** after the *n*-th consecutive collision of the same frame, choose a random integer K from 0 to $2^{\min(n,10)}$ - 1 and wait K x 512 bit times (slot time = 51.2 µs at 10 Mb/s) before retrying.
- After 16 failed attempts the frame is dropped and an error reported to the upper layer.

Why a 64-byte minimum frame?

A sender must still be transmitting when the collision signal returns from the farthest station, otherwise it will not know its frame was damaged. Hence *frame transmission time >= round-trip propagation delay (slot time)*. At 10 Mb/s, 512 bits (64 B) take 51.2 µs, which sets the maximum network diameter (about 2500 m for classic 10 Mb/s Ethernet). This is also why short payloads are padded to 46 B.

Why its importance is reduced in modern switched full-duplex Ethernet

- **Point-to-point links:** in a switched LAN each switch port connects to exactly one device through twisted pairs with separate transmit and receive pairs (or separate fibres). There is no shared medium to contend for.
- **Full duplex:** both ends can send and receive simultaneously, so there are no collisions and nothing to detect; CSMA/CD is **disabled** after auto-negotiation selects full duplex. The slot-time/minimum-frame constraint no longer limits cable length, only signal attenuation does.
- **Each switch port is its own collision domain**; the switch buffers frames and forwards them to the right port, so frames from different senders never collide on the wire (congestion shows up as buffering, drops or PAUSE frames instead).
- **Higher speeds:** 1000BASE-T supports half duplex in the standard but it is virtually never used; 10 Gb/s and faster Ethernet (802.3ae and later) defines **only full-duplex** operation, so CSMA/CD no longer exists there.
- **Effect:** near 100% link utilisation is possible (versus about 30-40% on loaded shared Ethernet) and latency is deterministic.

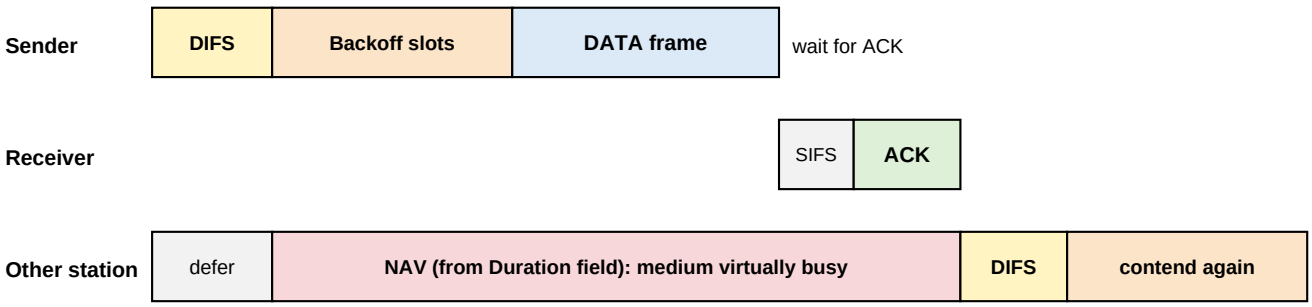
CSMA/CD is therefore kept in the standard for backward compatibility and appears mainly when a half-duplex device (old hub, duplex mismatch) is present. A *duplex mismatch* - one end full, the other half - is a classic fault that produces late collisions and poor throughput.

Q7. CSMA/CA in IEEE 802.11 and Comparison with CSMA/CD

Wi-Fi (IEEE 802.11) uses **CSMA/CA** as its Distributed Coordination Function (DCF). A wireless radio cannot detect a collision while it is transmitting: it cannot send and listen on the same channel at once, and the transmitted signal is far stronger than any signal arriving from another station. Because of the **hidden-node problem** (two stations out of range of each other but both in range of the AP) a sender also cannot be sure the receiver is free. So 802.11 *avoids* collisions and uses acknowledgements to detect the ones that happen anyway.

Operation (DCF)

- **Carrier sense:** physical sensing (clear channel assessment, CCA) plus **virtual carrier sense** using the Network Allocation Vector (NAV), set from the Duration field of frames overheard from others.
- If the medium is idle for a **DIFS** (e.g., 34 µs in 5 GHz OFDM), the station may transmit; if the medium is busy, it waits and then defers for DIFS plus a random **backoff**.
- **Backoff:** pick a random number of slots in [0, CW] (CWmin = 15 for OFDM PHYs, 31 for 802.11b); the counter decrements only while the medium is idle and freezes when it becomes busy. At zero the station transmits.
- **Acknowledgement:** the receiver sends an ACK after a short **SIFS** (16 µs) - SIFS < DIFS, so the ACK has priority over new contenders. No ACK = retry with the contention window doubled (up to CWmax = 1023).
- **Optional RTS/CTS:** for large frames the sender transmits RTS, the receiver answers CTS; both carry the Duration so all nodes in range of either set their NAV. This protects against hidden nodes.



Medium idle for DIFS -> random backoff (counts down only while idle) -> transmit -> receiver replies after SIFS.
No ACK before timeout = collision/loss: contention window doubles (15, 31, 63 ... 1023) and the frame is retried.

Figure 7.1 - CSMA/CA timeline: DIFS, random backoff, data, SIFS and ACK; other stations defer using NAV.

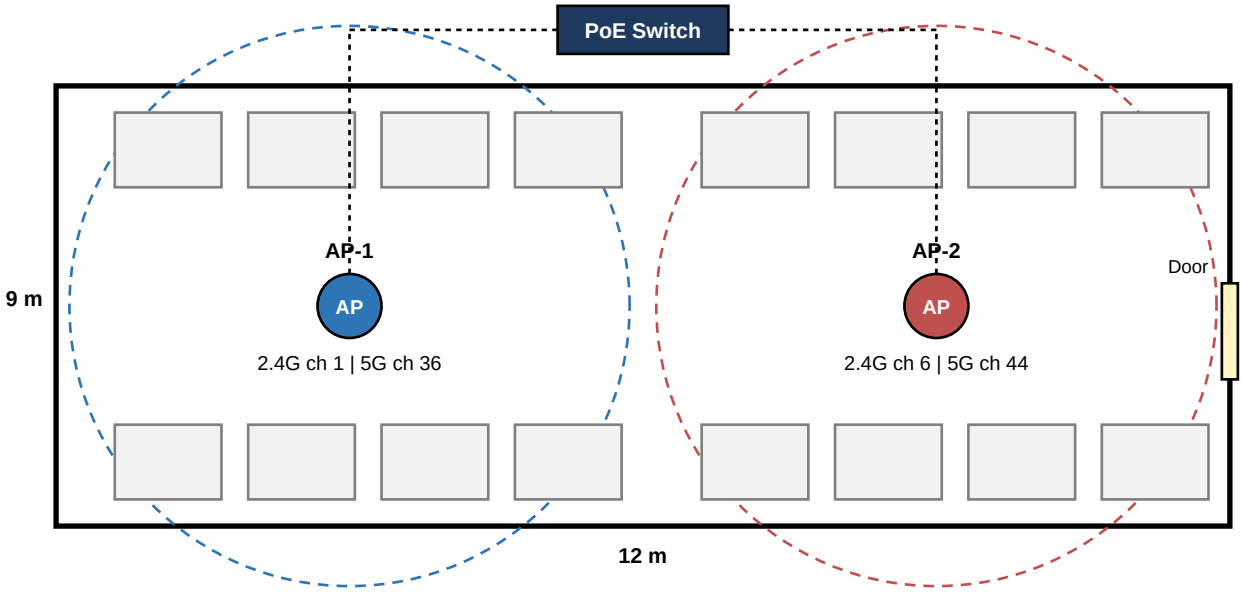
CSMA/CA vs CSMA/CD

Aspect	CSMA/CD (Ethernet 802.3)	CSMA/CA (Wi-Fi 802.11)
Objective	**Detect** collisions quickly and recover by aborting and retrying	**Avoid** collisions before they occur; recover by ACK-timeout and retransmission
Collision detection	Yes, while sending (wired medium, listen while talking)	Not possible (half-duplex radio, near-far effect, hidden nodes)
Medium	Wired shared bus / hub	Wireless shared radio channel
Before sending	Sense carrier; send immediately when idle	Sense carrier + wait DIFS + random backoff (+ optional RTS/CTS)
Acknowledgement	None at MAC (upper layers handle loss)	Mandatory MAC-level ACK for unicast frames
Backoff	Only after a collision (binary exponential)	Before every transmission and after failures (CW doubles)
Hidden node	Not applicable	Handled with RTS/CTS and NAV
Overhead / efficiency	Low overhead; efficient when load is low	High overhead (DIFS, backoff, SIFS, ACK); lower efficiency
Current use	Largely obsolete (full duplex switching)	Still fundamental to Wi-Fi (OFDMA in Wi-Fi 6 reduces contention)

Q8. Wireless LAN Design for a College Laboratory

Requirements and assumptions

- Computer laboratory of about **12 m x 9 m** (108 m^2), ceiling height 2.7 m, 40 student laptops/phones + 1 instructor PC, and some IoT/lab kits. Brick walls, one entrance door.
- Typical use: web browsing, cloud IDEs, SSH, software downloads, occasional video lectures - target >= 5 Mb/s per user at the same time.
- Wired Ethernet uplink with PoE available from an existing lab switch; internet via the college router/firewall.



Dashed circles: approximate coverage cell of each AP (about -67 dBm at the edge). Gray blocks: student workbenches (40 seats). Both APs are ceiling-mounted at 2.7 m and linked to the PoE switch by Cat6 (dashed lines); cells overlap slightly for roaming. Neighbouring labs should use 2.4 GHz channel 11 and 5 GHz channel 149.

Figure 8.1 - Proposed floor plan: two ceiling-mounted dual-band APs powered by a PoE switch.

Design specification

Item	Design decision	Reason
Access points	2 x dual-band Wi-Fi 6 (802.11ax) ceiling APs, PoE (802.3af), controller-less or cloud-managed	Capacity: about 20 active clients per AP is comfortable; coverage overlap for roaming
AP placement	At 1/4 and 3/4 of the room length (about 3 m and 9 m from the left wall), centred on width, ceiling mounted at 2.7 m, antenna facing down; away from metal cabinets, pillars and microwave ovens	Line of sight to every bench, minimal obstruction, equal distance, 15-20% cell overlap with signal at about -67 dBm at the edge
SSIDs	**CSLAB-Student** (students, WPA3/WPA2-Enterprise or WPA3-Personal); **CSLAB-Guest** (isolated, internet only, optional); SSID broadcast kept ON (hiding it gives no real security)	Meaningful name, same SSID on both APs so clients roam (ESS); separate SSID/VLAN for guests
2.4 GHz channels	AP-1: channel 1; AP-2: channel 6; adjacent labs: channel 11. 20 MHz width, TX power 8-11 dBm	Only 1, 6 and 11 are non-overlapping in the 2.4 GHz band; narrow width and low power reduce co-channel interference
5 GHz channels	AP-1: ch 36; AP-2: ch 44; adjacent labs: ch 149 (20 MHz, 40 MHz where interference is low); TX power 14-17 dBm; DFS channels avoided initially	Many non-overlapping channels, less interference, higher data rates; use channels permitted by the local regulator (WPC/DoT, India)
Band steering	Prefer 5 GHz for dual-band clients; keep 2.4 GHz for legacy and IoT; disable 802.11b rates (1, 2, 5.5, 11 Mb/s) and set minimum basic rate to 12 Mb/s	Improves airtime efficiency and reduces cell size
Addressing / VLAN	Student VLAN 20: **10.20.20.0/24**, gateway 10.20.20.1, DHCP pool 10.20.20.50-10.20.20.250 (8 h lease), DNS 10.20.0.10 / 8.8.8.8. APs on management VLAN 99: 10.20.99.2 and 10.20.99.3 (static). Guest VLAN 30: 10.20.30.0/24	254 usable addresses cover 40 clients + growth; DHCP simplifies client set-up; management traffic separated from users

Item	Design decision	Reason
Wired backbone	Cat6 from each AP to PoE switch (trunk carrying VLANs 20, 30, 99), uplink to the college core	Gigabit link per AP (multi-gig if Wi-Fi 6 AP supports it)
Security	WPA3-Personal (SAE) or **WPA2/WPA3-Enterprise with 802.1X/RADIUS** (individual credentials); AES-CCMP/GCMP only; disable WEP, TKIP, WPS; Protected Management Frames (802.11w) on; strong passphrase (>= 14 chars) if PSK; client isolation on guest SSID; firewall ACLs between VLANs; admin interface on HTTPS/SSH with non-default credentials and MFA; regular firmware updates; MAC filtering only as an inventory aid; rogue-AP detection and logging (syslog)	Protects confidentiality and integrity of the air interface; prevents unauthorised access and evil-twin attacks; complies with college IT policy and CERT-In guidance
Validation	Site survey (Ekahau / NetSpot / Wi-Fi Analyzer), iperf3 throughput test at corners and centre, check RSSI >= -67 dBm and SNR >= 25 dB, roaming test between APs	Confirm coverage and capacity before final installation

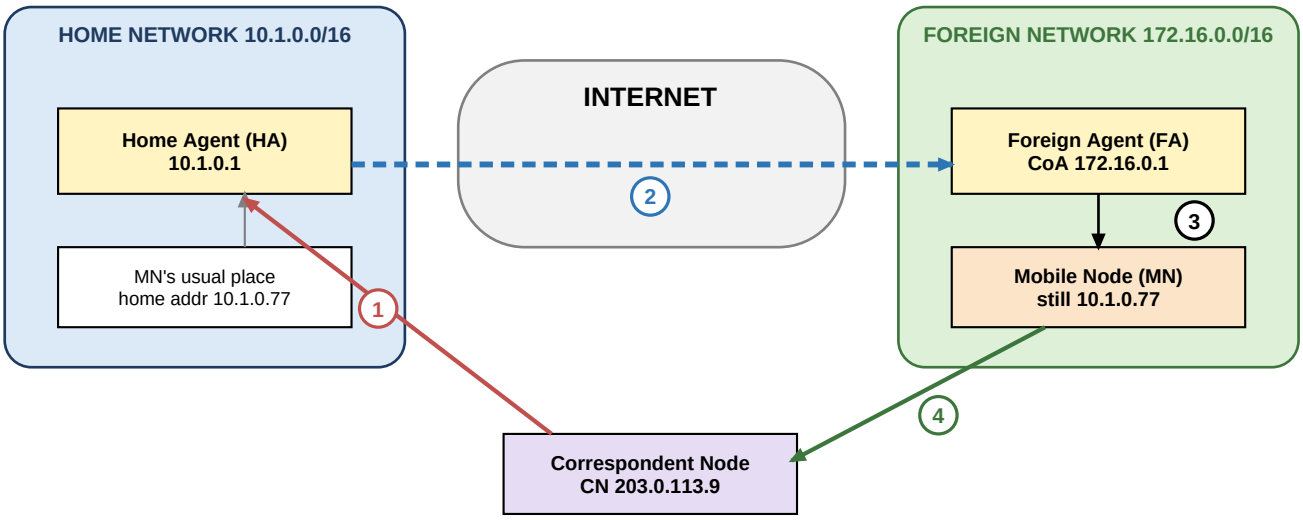
Q9. Mobile IP and the Need for Mobility Support

The problem

An IP address has two roles: it **identifies** a host and indicates its **location** (the network prefix tells routers where to forward). When a laptop or phone moves from its home network to another network it must obtain a new address there, but then (a) other hosts that know the old address can no longer reach it, and (b) established TCP connections, which are identified by the 4-tuple (source IP, source port, destination IP, destination port), break. Applications such as VoIP, video calls, SSH or downloads would have to restart at every handover. **Mobile IP** (RFC 5944 for IPv4; RFC 6275 for IPv6) lets a device keep a permanent address while changing its point of attachment, without changing routing in the whole Internet and without the use of special host routes.

Entities and terms

Term	Meaning
Mobile Node (MN)	The device that changes its point of attachment.
Home address	Permanent IP address of the MN in its home network; used by everyone to reach it.
Home Agent (HA)	Router on the home network. It receives packets addressed to the MN while it is away and tunnels them to the MN's current location; keeps the binding table.
Foreign Agent (FA)	Router on the visited network. Provides the care-of address, decapsulates tunnelled packets and delivers them to the MN.
Care-of Address (CoA)	Temporary address that reflects the current location: *foreign-agent CoA* (FA's address) or *co-located CoA* (an address obtained by the MN itself via DHCP).
Correspondent Node (CN)	Host communicating with the MN; needs no knowledge of mobility.



(1) CN sends to home address 10.1.0.77; the HA intercepts the packet (proxy ARP).

(2) HA encapsulates it (IP-in-IP, outer dst = CoA) and tunnels it across the Internet to the FA.

(3) FA decapsulates and delivers the original packet to the MN.

(4) MN replies directly to the CN (triangle routing); reverse tunnelling sends it via the HA instead.

Registration: MN -> FA -> HA (UDP port 434) binds 10.1.0.77 to CoA 172.16.0.1 for a limited lifetime.

Agent advertisements (ICMP router advertisement + mobility extension) tell the MN it has moved.

Figure 9.1 - Mobile IP operation: agent discovery, registration and tunnelling.

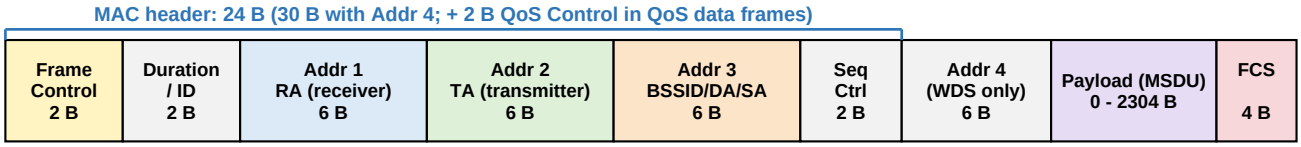
Operation in three phases

- **1. Agent discovery:** the HA/FA periodically broadcast *agent advertisements* (ICMP router advertisement with a mobility extension); the MN can also send agent solicitations. By comparing the advertised prefix, the MN detects whether it is at home or in a foreign network and obtains a CoA.
- **2. Registration:** the MN sends a *Registration Request* (UDP, port 434), via the FA, to the HA. The HA creates a **mobility binding** (home address -> CoA, lifetime) after authentication (HMAC/SPI) and replies with a Registration Reply. The HA also answers ARP on behalf of the MN (proxy ARP / gratuitous ARP).
- **3. Data delivery (tunnelling):** the CN sends to the MN's home address. The HA intercepts the packet, encapsulates it in a new IP header (IP-in-IP, protocol 4; outer source HA, outer destination CoA), and the FA decapsulates it and delivers it to the MN. Traffic from the MN to the CN goes directly (*triangle routing*) or is reverse-tunnelled through the HA if ingress filtering is in force.

Issues and improvements: triangle routing causes longer paths and extra latency; route optimisation (binding updates sent to CN) removes it in Mobile IPv6, which also eliminates the foreign agent (uses co-located addresses, IPv6 neighbour discovery and IPsec). Security of registration messages and handover delay are other concerns; later work such as Fast Handover and Proxy Mobile IPv6 improves them. Mobile IP complements, but is different from, Wi-Fi roaming (Layer 2 handover between APs in the same subnet), which does not need a new IP address.

Q10. Packet-Level Comparison: Wired Ethernet Frame vs Wireless LAN Frame

Both frames carry IP packets but differ because the wireless medium is shared, lossy and insecure by nature. The Ethernet frame is shown in Figure 2.1; the 802.11 data frame is in Figure 10.1.



Max frame 2346 B including FCS (before 802.11n aggregation). Addr 1 is always the immediate receiver.

Frame Control: Protocol ver (2b) | Type (2b) | Subtype (4b) | ToDS | FromDS | More Frag | Retry | PwrMgt | More Data | Protected | +HTC/Order.

Type: 00 management (beacon, probe, auth, assoc) | 01 control (RTS, CTS, ACK) | 10 data.

Figure 10.1 - IEEE 802.11 MAC frame format.

Documented captures

Wired (Ethernet II) - captured on the wired interface with a plain Wireshark capture (see Q3, frame 3). **Wireless (802.11)** - captured in *monitor mode* on Wi-Fi channel 6 (the radiotap pseudo-header adds RSSI, channel and data-rate information). Example commands on Linux:

```
sudo ip link set wlan0 down && sudo iw dev wlan0 set type monitor
sudo ip link set wlan0 up && sudo iw dev wlan0 set channel 6
wireshark -i wlan0 -k
# display filters:  wlan.fc.type_subtype == 0x08   (beacons)    wlan.addr == a4:5e:60:de:ad:01
```

No.	Source / TA	Destination / RA	Protocol	Info (Wireshark)
1	3c:84:6a:11:22:33 (AP)	ff:ff:ff:ff:ff:ff	802.11	Beacon frame, SN=1534, SSID=CSLAB-Student, channel 6
2	a4:5e:60:de:ad:01 (STA)	ff:ff:ff:ff:ff:ff	802.11	Probe Request, SSID=CSLAB-Student
3	a4:5e:60:de:ad:01	3c:84:6a:11:22:33	802.11	Authentication (Open System), then Association Request/Response
4	a4:5e:60:de:ad:01 (STA)	3c:84:6a:11:22:33 (AP)	802.11	QoS Data, ToDS=1, FromDS=0, Protected=1 (payload encrypted by CCMP)
5	-	a4:5e:60:de:ad:01 (RA only)	802.11	Acknowledgement, Flags=.....C (control frame, 14 B incl. FCS)

Figure 10.2 - Representative 802.11 monitor-mode capture (addresses are examples).

Decoded 802.11 QoS data frame (frame 4)

```
Radiotap Header: Channel 2437 MHz [BG 6], Signal -52 dBm, Data rate 130.0 Mb/s (MCS 15)
IEEE 802.11 QoS Data, Flags: .p.....T
    Type/Subtype: QoS Data (0x0028)
    Frame Control: 0x8841   (ToDS=1, FromDS=0, Protected=1)
    Duration: 44 microseconds
    Receiver address (Addr1 = BSSID): 3c:84:6a:11:22:33
    Transmitter address (Addr2 = SA) : a4:5e:60:de:ad:01
    Destination address (Addr3 = DA) : 00:1a:2b:3c:4d:5e   <- final wired destination (gateway)
    Sequence number: 2210       QoS Control: 0x0000 (TID 0, Best effort)
    CCMP parameters (PN, Key ID) .... Data (encrypted, 96 bytes)
    Frame check sequence: 0x1c9ae87b [correct]
```

Comparison table

Parameter	Wired Ethernet (802.3, Ethernet II)	Wireless LAN (802.11)
Medium access	CSMA/CD (half duplex) or none (full-duplex switched)	CSMA/CA (DCF) with ACK, optional RTS/CTS
Preamble / sync	7 B preamble + 1 B SFD	PLCP preamble + PLCP header (PHY layer, radio-dependent)
Header size	14 B (dst 6, src 6, type 2)	24 B (30 B with Addr4; +2 B QoS; +4 B HT control)

Parameter	Wired Ethernet (802.3, Ethernet II)	Wireless LAN (802.11)
Addresses per frame	2 (destination, source)	Up to 4: RA, TA, and DA/SA/BSSID depending on ToDS/FromDS
Frame types	Single frame type (data), plus control frames such as PAUSE	Management (beacon, probe, auth, assoc), Control (RTS, CTS, ACK, BlockAck) and Data
Control fields	Type/Length only	Frame Control (version, type, subtype, ToDS, FromDS, Retry, PwrMgt, Protected ...), Duration/ID, Sequence Control
Max payload	1500 B (MTU); jumbo frames up to about 9000 B	2304 B MSDU (A-MSDU up to 7935 B in 802.11n; larger with A-MPDU)
Min / max frame	64 B / 1518 B (1522 B with VLAN tag)	14 B ACK ... 2346 B (without aggregation)
Error handling	FCS (CRC-32); bad frames dropped; no retransmission at Layer 2	FCS (CRC-32) + mandatory ACK; retransmission with Retry bit; fragmentation optional
Reliability	Very reliable medium; BER about 10^{-12}	Lossy medium; interference, fading, hidden nodes; dynamic rate adaptation
Security	None at Layer 2 by default (802.1X / MACsec optional)	Built-in: WPA2/WPA3 (CCMP/GCMP); the 'Protected' bit marks encrypted payloads
Overhead	Low: header + FCS 18 B, IFG 12 B	High: PLCP preamble, 28-36 B MAC overhead, SIFS, ACK, backoff
Visibility in capture	IP and upper layers visible directly	Needs monitor mode (radiotap header); data encrypted unless the Wi-Fi key and the 4-way handshake are available
Duplex	Full duplex possible (separate pairs)	Half duplex (one channel shared in time)
Mobility	Fixed cable	Stations roam between APs (association, reassociation)

References

[1] IEEE Std 802.3-2022, *IEEE Standard for Ethernet*. IEEE.

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[6] D. Plummer, RFC 826: *An Ethernet Address Resolution Protocol*, IETF, 1982.

[7] J. Kurose and K. Ross, *Computer Networking: A Top-Down Approach*, Pearson (Chapter 6: The Link Layer and LANs).

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[10] Wireshark Foundation, *Wireshark User's Guide* and sample captures, https://www.wireshark.org/docs/ and https://wiki.wireshark.org/SampleCaptures.

Repository structure (see README.md)

```
pccst501-assignment3/
|-- README.md
|-- report/PCCST501_Assignment3_Report.pdf
|-- captures/        (ethernet_capture.pcapng, wlan_capture.pcapng)
|-- screenshots/     (q3_ethernet_frames.png, q4_mac_table.png, q10_wlan_frame.png)
|-- diagrams/        (ethernet_frame, wlan_design, mobile_ip)
`-- docs/            (notes, references)
```